Fact-based News Verification System

A Hybrid System with BERT Classification, GNews Evidence Retrieval, and Fine-tuned NLI Verification

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Abstract: This project develops a fact-based news verification system for online news articles. Instead of relying only on article-level fake/real classification, the system combines a fine-tuned BERT baseline classifier with sentence-level evidence retrieval and a fine-tuned DeBERTa-based Natural Language Inference verifier. The input article is cleaned and split into sentence-level verification units. Each sentence is used to retrieve external evidence from GNews, and the NLI model determines whether the evidence supports, refutes, or provides insufficient information for the sentence. The sentence-level results are aggregated into an article-level verdict. The system also includes sentiment analysis, LDA-based topic analysis, and a Streamlit chat interface to improve interpretability and user interaction.

Keywords: BERT Classification, GNews Retrieval, Natural Language Inference, Evidence Verification, Sentiment Analysis, Topic Modelling, Streamlit Chat

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1 Business Problem

Online news spreads quickly, but ordinary readers rarely have time to check whether the factual statements in an article are actually supported by external evidence. This creates a practical verification problem rather than only a text classification problem. A news story may look professional, use neutral tone, and resemble real reporting style while still containing unsupported, outdated, or false factual statements.

This motivates a shift away from treating the task as simple fake/real classification. Three problems are especially important:

- article-level fake/real classifiers cannot explain which sentence or claim is false;
- style-based classifiers may learn topic bias, source bias, or writing patterns instead of factual correctness;
- real-world articles often mix true, false, and unverifiable statements in the same document.

The goal of this project is therefore to build an explainable first-pass verification tool that retrieves external evidence and verifies sentence-level factual statements. The system is designed to:

- classify the overall article using a BERT baseline model;
- split the article into sentence-level verification units;
- retrieve external evidence from GNews for each sentence;
- use a fine-tuned NLI model to classify evidence-sentence pairs as supported, refuted, or not enough information (NEI);
- aggregate sentence-level results into an article-level verdict;
- provide sentiment, topic, and chat-based interaction features for interpretability and usability.

The intended users are students, researchers, journalists, and general readers who want a transparent way to inspect whether a news article is supported by other reporting. The system is not a replacement for professional fact-checking. Instead, it serves as an explainable verification assistant that combines fast screening with evidence-based semantic reasoning.

2 Datasets Used

The final report narrative separates the data sources by their role in the pipeline rather than treating everything as one generic dataset. The system uses local fake/real datasets for baseline classification, news-related claim-evidence data for NLI training, GNews as a live evidence source at inference time, and the fake/real datasets again for topic exploration.

2.1 Baseline Fake/Real Classification Dataset

The local repository uses four FakeNewsNet-style CSV files for the baseline BERT classifier. These files mainly provide title-level text rather than full article bodies, so the baseline model should be described honestly as a title or headline classifier with limited article context.

These files are suitable for a baseline classifier because they provide labeled examples of fake and real news headlines. However, because they do not mainly contain full article bodies, they are insufficient for sentence-level factual verification on their own.

2.2 NLI Training Dataset

For the NLI verifier, the training data is organized as claim-evidence-label triples. Each example pairs a factual statement with a supporting, refuting, or insufficient evidence passage. This training format is closer to the actual verification task than binary fake/real classification.

Table 1: Local fake/real datasets used for baseline classification

File	Rows	Label	Role
gossipcop_fake.csv	5,323	Fake	Title-level fake-news examples for BERT training.
gossipcop_real.csv	16,817	Real	Title-level real-news examples for BERT training.
politifact_fake.csv	432	Fake	Additional fake-news titles for domain diversity.
politifact_real.csv	624	Real	Additional real-news titles for domain diversity.
Total	**23,196**	–	Local baseline data with 17,441 real and 5,755 fake rows.

Table 2: Training format for the NLI verifier

Field	Meaning
`claim`	Sentence or factual statement to be verified.
`evidence`	Retrieved or curated news evidence passage.
`label`	Verification target: `supported`, `refuted`, or `NEI`.

The verifier is trained on news-related fact-checking or stance-verification data, where evidence-claim pairs are mapped into supported, refuted, and NEI labels. When stance-style datasets are used, a practical label mapping is:

- `agree` → `supported`;
- `disagree` → `refuted`;
- `discuss` or `unrelated` → `NEI`.

This mapping is useful but imperfect, because stance labels are not identical to factual entailment labels. For that reason, manual inspection, filtering, or a small manually labeled test set is still needed before final evaluation.

2.3 Live Evidence Source

GNews is not treated as a training dataset. Instead, it is used as a live external evidence source at inference time. For each sentence-level verification unit, the system queries GNews and collects returned article metadata such as title, description, content snippet, source, publication date, and URL.

This distinction matters for the report narrative: the baseline classifier and NLI verifier are trained offline, while GNews provides current evidence during runtime. The retrieval step therefore connects the static models with up-to-date news reporting.

2.4 Topic Analysis Dataset

The same local GossipCop and PolitiFact data are also used for LDA-based topic analysis. In this module, the labels are not used for verification decisions. Instead, the documents are vectorized to explore recurring themes, topic-word distributions, and fake/real topic patterns in the training corpus.

3 Solution Approach

The final system should be described as an evidence-based news verification pipeline rather than as a simple fake-news classifier. The baseline BERT model remains important, but it now serves as an article-level prior alongside a sentence-level retrieval and NLI verification pipeline.

3.1 System Overview

The end-to-end flow of the system is shown below:

```
Article Input → Text Cleaning → BERT Baseline Classifier
→ Sentence Splitting → GNews Retrieval → Evidence Reranking
→ Fine-tuned DeBERTa-NLI Verifier → Article-level Aggregation
→ Final Verdict + Evidence + Sentiment + Topic + Chat
```

This architecture balances speed and interpretability. The BERT classifier gives an immediate article-level signal, while the retrieval and NLI stages explain which specific factual statements are supported, contradicted, or still unverified.

3.2 Text Cleaning

Before classification and retrieval, the input article is normalized to reduce noise. The cleaning stage removes URLs, strips social-media artefacts, and normalizes whitespace and punctuation. This improves consistency for both the baseline classifier and the retrieval module.

The cleaning step is deliberately lightweight. Its goal is not to rewrite the article, but to preserve the original wording as much as possible while making the input safe and stable for downstream processing.

3.3 BERT Baseline Classifier

The baseline model uses `bert-base-uncased` fine-tuned for binary fake/real classification. Because the available local data are mainly title-level records, the classifier should be presented as a fast article-level prior rather than as a complete factual verifier.

This distinction is important for the report. The BERT classifier does not determine whether the article is true or false in a strict fact-checking sense. Instead, it estimates whether the article resembles fake or real examples in the training data. Its output is later combined with evidence-based verification signals rather than used as the only decision criterion.

3.4 Sentence-level GNews Evidence Retrieval

Earlier experiments with claim extraction showed that aggressive claim simplification often reduced search quality. Important context was sometimes lost, making the search query too vague or too distorted. Therefore, the final system uses sentence-level retrieval units instead of heavily simplified claims.

Each sentence is preserved as much as possible and submitted to GNews after query-safe sanitization. The retrieval logic focuses on:

- punctuation removal and cleanup that do not change the core wording;
- query length control for API-safe search strings;
- preserving original word order whenever possible;
- searching over returned title, description, and content fields;
- reranking evidence based on lexical overlap, entities, numbers, and source relevance.

This design improves explainability because the system can show users exactly which sentence triggered which retrieval query and which evidence was selected for verification.

3.5 Fine-tuned DeBERTa-NLI Verification

The central upgrade in the final system is the move from heuristic overlap matching to a fine-tuned Natural Language Inference verifier. The NLI model receives the retrieved evidence as the premise and the article sentence as the hypothesis:

- **Premise**: retrieved evidence snippet;
- **Hypothesis**: sentence from the input article.

The NLI output is mapped into verification labels as follows:

- entailment → supported;
- contradiction → refuted;
- neutral → NEI.

To satisfy the deep-learning requirement and improve semantic reasoning, the verifier is initialized from a DeBERTa-based NLI checkpoint and then fine-tuned on news-related evidence-claim pairs. This is more robust than heuristic token matching because it can model contradiction, paraphrase, and relevance at the sentence-pair level.

3.6 Article-level Aggregation

The system aggregates sentence-level verification outputs into a final article-level verdict. A practical aggregation rule is:

- many supported sentences → `Likely True`;
- many refuted sentences → `Likely False`;
- mostly NEI results or weak evidence → `Unverified`.

The BERT baseline is used as an additional article-level signal rather than as a hard override. This hybrid design allows the final verdict to reflect both the training-data prior and the retrieved evidence trail.

3.7 Sentiment Analysis

Sentiment analysis is included as an interpretability feature rather than a truth-verification signal. The system uses a transformer sentiment model from the CardiffNLP sentiment family, with a lighter fallback such as VADER when necessary. The output labels are positive, neutral, and negative.

This module helps users understand the tone of the submitted article or the tone of retrieved reporting. However, sentiment does not determine whether a story is true or false. A false article can sound neutral, and a true article can be emotionally framed, so sentiment is used only to improve interpretability.

3.8 LDA Topic Analysis

The topic analysis component uses `CountVectorizer` and Latent Dirichlet Allocation (LDA) to explore patterns in the local training corpus. The module can show top words per topic, sample documents, document-topic probabilities, and fake/real distributions by topic.

This analysis is useful for exploratory understanding of the dataset and for showing how different topics appear in fake and real subsets. It is not part of the final truth-verification decision and should be described as an interpretability and exploratory-analysis feature.

3.9 Chat Interface

The Streamlit chat tab provides a conversational interface over the system's functions. Through chat-style prompts, the user can request article verification, inspect sentence-level search units, view sentiment output, and obtain short article previews.

To keep the report accurate, the chat interface should be described honestly as a rule-based interaction layer built with Streamlit chat components. It maps user keywords or menu choices to existing verification functions. It is not an open-ended large-language-model chatbot.

4 Test Results

The evaluation section should reflect the upgraded system design. Instead of showing only one fake/real accuracy number, the report now separates baseline classification, NLI verification, and end-to-end demonstration evidence.

4.1 BERT Baseline Evaluation

The BERT baseline should still be evaluated with standard binary classification metrics. Because the local data are imbalanced and title-based, the table below is presented as the required reporting format for the final submission.

Table 3: Baseline BERT classification results

Model	**Accuracy**	**Precision**	**Recall**	**F1**
BERT baseline classifier	Pending final run	Pending final run	Pending final run	Pending final run

The final submission should also include a confusion matrix and a short note on the limitation of title-level training data. This makes clear that the baseline is a useful screening component, but not the main factual verification module.

4.2 NLI Verifier Evaluation

The NLI verifier should be evaluated with three-way classification metrics, including accuracy, macro-F1, and class-wise F1 for supported, refuted, and NEI. A direct comparison against the earlier heuristic verifier is especially important because it demonstrates the benefit of semantic inference.

Table 4: Verifier comparison table for the final report

Model	**Accuracy**	**Macro-F1**	**Supported F1**	**Refuted F1**	**NEI F1**
Heuristic verifier	Pending final run	Pending final run	Pending final run	Pending final run	Pending final run
Fine-tuned DeBERTa-NLI	Pending final run	Pending final run	Pending final run	Pending final run	Pending final run

The final version of this subsection should also include a confusion matrix and brief discussion of where the NLI verifier improves on heuristic matching, especially for paraphrases, negation, and sentences that contain distracting numbers.

4.3 End-to-end Verification Case Study

A complete evaluation should also include one worked example showing how the system behaves on a real article. Table 5 gives a suitable format for documenting sentence-level evidence, predicted labels, and the type of improvement expected from the NLI verifier.

Table 5: Representative end-to-end case-study structure

Sentence	**Top Evidence**	**Target Label**	**Comment**
Brockman disclosed financial ties and a nearly $30B stake.	Reuters	Supported	Strong evidence match for the financial-interest statement.
Musk filed a lawsuit over the for-profit versus nonprofit issue.	Economic Times	Supported	Retrieved article directly addresses the litigation context.
Brockman's independence may be compromised by startup investments.	Rappler	Supported	Evidence is related and provides supporting discussion rather than exact lexical overlap.
Stakes in startups and Altman's family fund raise conflict-of-interest concerns.	Economic Times	Supported	Useful example of multi-entity matching across one sentence.
The trial and ChatGPT launch happened in 2022.	MarketScreener	NEI → Supported	Heuristic matching can overreact to unrelated numbers; NLI is intended to reduce this error.

This kind of case study is important because it demonstrates not only whether the final verdict is plausible, but also whether the evidence trail shown to the user is understandable and trustworthy.

4.4 Error Analysis and Discussion

Even after the NLI upgrade, several error sources remain important:

- **Retrieval errors**: GNews may return related but still insufficient evidence.
- **Snippet limitation**: the returned content can be truncated, which weakens premise quality for NLI.
- **NLI ambiguity**: long or multi-fact sentences can confuse the verifier because one sentence may contain both supported and unsupported subclaims.
- **Label-mapping noise**: stance datasets do not perfectly match factual entailment labels.
- **Aggregation sensitivity**: a single wrongly refuted sentence may affect the article-level verdict disproportionately.

Including this discussion makes the report more realistic and shows that the team understands the limitations of evidence-based verification in practical deployment settings.

5 Limitations and Future Work

The current system is more transparent than a pure article-level classifier, but it still depends heavily on retrieval quality and dataset fit. If the retrieved news snippets are incomplete or weakly related, even a strong NLI model cannot produce reliable verification labels. In addition, the baseline fake/real data are title-heavy, so they are better suited to screening than to deep factual reasoning.

Future work should focus on four areas. First, the team should expand the NLI training set with more news-specific claim-evidence pairs and a cleaner manual validation split. Second, the retrieval stage can be improved with better sentence selection, reranking, and source filtering. Third, the aggregation rule can be calibrated with confidence weighting rather than simple counting. Fourth, the system can be extended to support multi-hop evidence and longer article-body verification beyond sentence-level snippets.

6 Conclusion

The final system demonstrates a hybrid approach to news verification. The BERT classifier provides a fast article-level fake/real prior, while sentence-level GNews retrieval and the fine-tuned NLI verifier provide evidence-based semantic verification. Sentiment analysis, LDA topic modelling, and the chat interface improve interpretability and usability.

Compared with a pure fake-news classifier, this design is more transparent because it shows which sentences are supported, refuted, or still unverified and which external evidence was used. At the same time, the system remains limited by retrieval quality, snippet completeness, label-mapping noise, and the final accuracy of the NLI verifier. Even with those limitations, the project shows why evidence-based verification is a more convincing direction than relying only on article-level style classification.

7 References

7.1 Libraries and Frameworks

1. **Streamlit**. Used for the interactive application interface. Official documentation: Streamlit documentation. Chat component reference: Streamlit chat API reference. Public source code: streamlit/streamlit.
2. **PyTorch**. Used as the deep learning runtime for BERT and DeBERTa training and inference. Official documentation: PyTorch documentation. Public source code: pytorch/pytorch.
3. **Hugging Face Transformers and Datasets**. Used for model loading, tokenization, fine-tuning, and dataset preparation. Official documentation: Transformers documentation and Datasets documentation. Public source code: huggingface/transformers and huggingface/datasets.
4. **scikit-learn**. Used for train-test splitting, evaluation metrics, `CountVectorizer`, and LDA topic modelling. Official user guide: scikit-learn user guide. LDA reference: LatentDirichletAllocation documentation. Public source code: scikit-learn/scikit-learn.
5. **pandas, NumPy, and Requests**. Used for CSV processing, numerical operations, and API calls. Official documentation: pandas, NumPy, and Requests.

7.2 Models, APIs, and Data Sources

1. **`bert-base-uncased`**. Pretrained encoder used for the baseline fake/real classifier. Model card: Hugging Face model card.
2. **DeBERTa / DeBERTa-v3**. Backbone used for semantic verification through NLI fine-tuning.

Model resources: DeBERTa-v3 base model card.

3. `MoritzLaurer/deberta-v3-base-mnli-fever-anli`. Practical starting checkpoint for NLI-based verification. Model card: Hugging Face model card.
4. **News-related NLI or stance-verification data**. Used to build claim-evidence-label training pairs for the verifier. One public reference is the Fake News Challenge Stage 1 dataset: FNC-1 repository. The report should note that stance labels require mapping and manual inspection before use as NLI labels.
5. **GNews API**. Live evidence source used at inference time for sentence-level retrieval. Official documentation: GNews API documentation.
6. `cardiffnlp/twitter-roberta-base-sentiment-latest`. Sentiment model used as an interpretability feature. Model card: Hugging Face model card.
7. **FakeNewsNet-style GossipCop and PolitiFact data**. Local fake/real datasets used for the baseline classifier and topic analysis. Public repository reference: KaiDMML/FakeNewsNet.

8 Individual Contributions

Table 6: Planned contribution summary for the final report

Team Member	**Main Responsibility**	**Contribution Details**
Sun Yuchen	Dataset preparation, baseline BERT training, evaluation, and topic analysis	Prepared the fake/real datasets, organized the baseline training workflow, documented classifier evaluation, and supported LDA-based dataset exploration.
Zhao Jiahui	GNews retrieval, sentence-level pipeline, Streamlit app, and chat interaction	Implemented retrieval logic, sentence-level verification flow, evidence display, and the user-facing interface for verification and chat-style interaction.
Zhang Yuxuan	NLI dataset construction, DeBERTa-NLI fine-tuning, sentiment module, and verifier evaluation	Prepared claim-evidence training pairs, fine-tuned the NLI verifier, integrated sentiment analysis, and designed the verifier comparison metrics.
Shared	Report writing, demo preparation, and error analysis	All team members contributed to report revision, final presentation materials, discussion of limitations, and interpretation of system results.

9 GenAI / LLM Declaration

Generative AI tools were used to assist with debugging, report structuring, explanation refinement, and code review. The team manually reviewed, modified, and validated the generated suggestions before including them in the project. No confidential or private user data was submitted. The final system design, implementation decisions, experiments, and conclusions remain the responsibility of the project team.